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A smart mirror to promote a healthy lifestyle

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Highlights

- Wize Mirror seeks to move semeiotic analysis of the face to normal life settings.
- The goal is to promote primary prevention of cardio-metabolic diseases.
- Wize Mirror will support people in self-monitoring and self-assessing wellbeing.
- It is non-intrusive, interacts natural with user, and gives tailored guidance.
- The guidance will give with coaching and ensure awareness and comprehension.

ICT solutions to foster behavioural change have been shown to be effective in implementing primary prevention in terms of a healthy lifestyle. Primary prevention is the most viable approach to reduce the socio-economic burden of chronic and widespread diseases, such as cardiovascular and metabolic diseases. In this paper, we present a novel multisensory device, the Wize Mirror, which is under development in the EU FP7 Project SEMEOTICONS. The Wize Mirror detects and monitors over time semeiotic face signs related to cardio-metabolic risk, and encourages users to reduce their risk by improving their lifestyle.

Introduction

It is well-known that prevention is the best strategy to limit the spread of cardio-metabolic diseases, which are the leading cause of mortality worldwide (Mathers, Fat, & Boerma, 2008). Indeed, cardio-metabolic risk factors are mainly related to modifiable factors pertaining to people's lifestyle, such as dietary habits, physical activity, tobacco and alcohol use, stress and psychological conditions. New perspectives on primary prevention rely on the *empowerment* of individuals, in terms of their ability to self-monitor their health status and act upon their lifestyle. Nowadays people are aware of the importance of a correct lifestyle, and they want to be active actors in the acquisition and maintenance of wellbeing.

Over the last few years, supportive technological instruments have invaded the market, mainly in the form of wearable devices (wristbands, smartwatches, eyewear, wearable bio-monitors) and applications on smart devices (such as Runtastic or Melarossa). According to an Internet-based survey of thousands of Americans (Ledger & McCaffrey, 2014), in September 2013 one in ten US consumers over the age of 18 owned a modern

activity tracker. Interestingly, there was a bimodal distribution of users by age: the adopters are divided into youngsters (25–34 years), who focus on fitness optimisation, and older users (55–64), who focus on improving overall health and extending their lives. But what contrasts with the increasing adoption of wearables is that the majority of consumers stop using the device within six months of receiving it. In other words, most of the devices on the market fail to drive long-term, sustained engagement. As a consequence, they fail to make a long-term impact on their users' health.

We believe the key to success is sustained engagement, based on the promotion of behaviour change towards *wellness* as a whole and long-term health objectives. Enhancing wellness is an effective way to promote participation and motivate people to change their habits. In a year-long research study from the University of Michigan (Segar, Eccles, & Richardson, 2011), participants who adhered to a long-time exercise regime did not want to lose weight, but rather to “enhance their daily life and wellness”.

This is the direction in which the European project SEMEOTICONS (“SEMEiotic Oriented Technology for Individual's CardioMetabolic risk self-assessment and Self-monitoring”) is moving (SEMEOTICONS FP7 European Project, 2013). SEMEOTICONS started in November 2013 to develop a multisensory device in the form of a mirror, called the “Wize Mirror”, which comfortably fits the home as a piece of house-ware, but also pharmacies and fitness centres. By analysing data acquired unobtrusively via a suite of contactless sensors, the Wize Mirror detects on a regular basis any physiological change relevant to cardio-metabolic risk factors. The computation and delivery of a comprehensive Wellness Index enables individuals to estimate and track over time their health status and their cardio-metabolic risk. Finally, the Wize Mirror offers personalised guidance towards the achievement of a correct lifestyle, via tailored coaching messages.

The Wize Mirror is designed to meet a two-fold objective: stimulating *initial adoption and utilisation*, by providing a pleasant user experience; and supporting *long-term engagement*, by helping people to establish new positive habits. To this aim, the main features of the Wize Mirror are: the provision of day-by-day monitoring in an unobtrusive way; the automatic assessment of physiological conditions via advanced data processing algorithms; and the promotion of sustained behaviour change towards long-term wellness objectives. These functionalities are developed by taking into account theories belonging to different disciplines (psychology, motivation and communication science, social marketing, behavioural theories and economics).

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Section snippets

The Wize Mirror at a glance

The inspiring idea behind the Wize Mirror is that the face is the preeminent channel of communication among humans: it is a mirror of status, emotions and mood. As such it is the base of *medical semeiotics*, which reveals the healthy status of an individual through a combination of physical face signs (e.g., skin colour, subcutaneous fat), and facial expressions. Dating back as far as Aristotle, medical semeiotics is still used today by medical doctors. The Wize Mirror moves medical semeiotics ...

State of the art

Over the last decades, the growing demand for telemedicine and health care has been met by many systems for automating medical data collection and diagnosis, monitoring physiological conditions, and performing diagnosis and treatment. In most cases, these systems are equipped with sophisticated and expensive medical machinery (like electrocardiography (ECG) or electroencephalography (EEG)) and they are not intended for everyday residential use.

One of the emerging solutions for the integration ...

Touchless data acquisition

The Wize Mirror features an advanced sensing framework for unobtrusive acquisition of videos, images, and 3D scans of individuals standing in front of the mirror. The imaging devices include a visible camera, a multispectral imaging system, and 3D optical sensors. The platform also includes a device for breath analysis (Fig.2).

Visible cameras are used for emotional analysis: video sequences are captured to look for signs of stress, fatigue and anxiety. The prototype under development includes ...

Multimodal data processing

The multimodal data acquired by the sensors are processed to extract reliable computational measures correlated with clinical risk factors.

The clinical partners in SEMEOTICONS have produced a *semeiotic model* of the face for cardio-metabolic risk (Coppini, Favilla, Gastaldelli, Colantonio, & Marraccini, 2014), defining cardio-metabolic risk factors and the set of parameters that the Wize Mirror can produce as surrogate risk indicators. Table 1 shows the risk factors (which include physical, ...

Data fusion and synthesis

The WI is a composite measure of the health status of an individual, which is traced over time and reported in a health diary for the user to consult.

There are many definitions of wellbeing indices in the literature, though there is no standard measurement of the health status of individuals or population groups. The health status may be measured either by an observer (e.g., a physician), or by asking people to report their health perception (about physical functioning, emotional well-being, ...

User profiling

The Wize Mirror is designed to ensure a comprehensive and long-lasting approach to lifestyle changes. The main requirement is that the support is strongly tailored. Tailored information has been proved to be effective in giving consumer information and is generally preferred by patients. Tailoring means to reach one specific person based on characteristics that are unique to that person, related to the outcome of interest and derived from an individual assessment. Therefore, *profiling* the ...

Personalised guidance

The final goal of the Wize Mirror is to make people active actors in their health care. Decision support systems are gaining increasing attention as a solution, to interpret data collected by tele-monitoring systems and to stimulate people-changing behaviour (Oinas-Kukkonen, 2010). Customised remote systems that monitor and assist users are usually called Personalised Guidance Systems (PGS) or Personalised Support Systems (PSS). Often, such systems are internet-, mobile-, or game-based (Honka ...

System validation

The Wize Mirror is being realised in a cycle of development, test and validation phases. The cycle ends with a final verification of the ability of the system to estimate cardio-metabolic risk. The validation is based on a study population consisting of two groups:

the *Reference Dataset* (Section 8.1), for the intermediate validation of individual algorithms, and the *Clinical Validation Dataset* (Section 8.2), for the study of the reproducibility of measurements and final clinical validation. The ...

Discussion and conclusions

We have presented the Wize Mirror, a platform for the estimation of cardio-metabolic risk from sensed facial data and the delivery of personalised user guidance towards lifestyle change. The platform is under development in the EU FP7 Project SEMEOTICONS. Building the Wize Mirror implies responding to several challenges: *technological*, *scientific*, and *clinical* challenges.

The *technological* challenge is the development of a non-intrusive platform, seamlessly integrated into daily-life ...

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2020, Expert Systems with Applications

Citation Excerpt :

...These two camera setups tend to be used more for applications (see Section 4) like human action recognition, or fall prevention and detection: the first (egovision) can detect the wearer's motion patterns with respect to the environment, as well as activities involving the hands and handled objects; whereas the second can recognise motion patterns involving the full body. In some medical applications of computer vision or methods relying on face analysis, camera setup might need to follow a specific or bespoke setup (so that the sensor is closer to the analysed body part), as in Huimin et al. (2017), Lewis, Davila, and Porges (2018), Li, Yadollahi, and Taati (2017) and Maclaren, Aksoy, and Bammer (2015), or be disguised in an everyday item such as a mirror (Andreu et al., 2016; Colantonio, Coppini et al., 2015; Henriquez et al., 2017). Using RGB-only devices Moved by the scarcity of videos, and the small size of datasets for action recognition, Carreira and Zisserman (2017) propose a new dataset, namely the Kinectics Human Action Video (KHAV or simply Kinectics) dataset....

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